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ABSTRACT

Classical molecular dynamics simulations of hydrogen plasma, employing quantum pseudopotentials such as the Kelbg pseudopotential, provide a computationally efficient alternative to fully quantum methods. This approach avoids the high cost of *ab initio* quantum simulations while enabling the study of dynamic properties and time evolution in dense partially ionized plasmas. However, classical methods often fail to properly account for the quantum exchange correlations that govern the behavior of electrons. In this work, we systematically investigate the stability of classical hydrogen plasma using the standard and improved Kelbg pseudopotentials and molecular dynamics across the range of coupling parameters $\Gamma = 0.25$ – 2.5 . We demonstrate that both pseudopotentials lead to unphysical clustering of electrons with identical spin projections at $\Gamma > 0.5$, resulting in a complete system collapse at $\Gamma \geq 1.0$. This artifact, clearly visible in radial distribution functions and energy evolution, manifests itself as an unphysical peak at distances below the thermal de Broglie wavelength, reflecting the failure of the pairwise pseudopotential to reproduce fermionic exchange effects. Our analysis establishes $\Gamma \approx 0.5$ as the upper limit of applicability for these classical approaches. The results highlight a critical gap in current methodologies and provide a clear motivation for the development of physically justified corrections that explicitly emulate quantum exchange effects.

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I. INTRODUCTION

It is well-known that a classical system of point charges is unstable thanks to the famous Earnshaw's theorem. This fact demonstrates that quantum effects play a crucial role for the very existence of plasma even in the case of negligible interparticle interaction. On the other hand, quantum effects are very difficult to treat both theoretically and numerically. This contradiction leads to numerous approximate approaches with unknown accuracy.

Theoretical investigation of a classical two-component plasma (TCP) was originated in the famous work by Debye and Hückel.¹ A hundred years later, the screening Debye potential remains one of the few exact results for this system in the limit of weak coupling. The Debye–Hückel expression was extended in the virial expansion for the fully ionized TCP free energy by Vedenov and Larkin.² A diagram technique and the Beth–Uhlenbeck theory were used to obtain two additional terms in comparison with the Debye–Hückel theory. No significant theoretical results for the classical TCP have been revealed since that time. This fact can be explained by the complexity of the system caused by the formation of bound states (atoms, ions, and

molecules). Much effort was spent on the problem of the divergence of the statistical sum of an atom.^{3–5} However, the most fundamental question is connected with the stability of TCP. This problem was solved in a series of works by Dyson and Lennard and was excellently reviewed by Lieb.⁶ It was proved that two conditions should be satisfied in order to prevent the collapse of TCP: (i) the uncertainty principle and (ii) the Pauli principle. The last condition means that at least one type of particle in the TCP should be fermions.

Thus, any theoretical investigation or numerical simulation of TCP should take into account these two main quantum principles. As theoretical consideration is currently possible only at very weak coupling, atomistic simulation with Monte Carlo or molecular dynamics methods has become the dominating approach. To avoid instabilities during a classical simulation, the interparticle interaction in TCP should be somehow modified. For example, naive methods include cutting off the electron–ion potential⁷ or a smooth decay to a finite value.⁸ A more rigorous way, taking into account the uncertainty principle, is the derivation of a pair pseudopotential from the diagonal part of a two-particle density matrix or, in other words, from the

Slater sum.⁹ A pseudopotential is usually temperature-dependent and takes into account the uncertainty principle, which is expressed as a finite value of the electron-ion pseudopotential at zero distance. There are many examples of pseudopotentials, but the most commonly used are those derived by Kelbg,¹⁰ Barker,¹¹ and Deutsch.¹² The latter work provides a detailed analysis of temperature-dependent Coulomb interactions in hydrogenic systems. All these pseudopotentials are intended to be applied at relatively high temperatures of the order of Rydberg energy ($Ry \approx 13.6$ eV). To avoid unphysical effects at lower temperatures, a temperature-dependent parameter is usually introduced into a pseudopotential.^{13,14} In Ref. 14, the so-called improved Kelbg pseudopotential was offered, which allowed to make simulations at rather low temperatures (0.2 Ry).

The Pauli principle should be taken into account by antisymmetrization of the many-particle density matrix of a system. In practice, a two-particle density matrix is usually used instead. A general expression for the density matrix in the first order of perturbation theory was obtained by Kelbg.¹⁰ Singlet and triplet states of a pair of electrons can be taken into account by symmetrization and antisymmetrization of two-particle density matrices.^{12,14–16} Using this procedure, one can obtain pseudopotentials for indistinguishable particles that approximately incorporate exchange effects. The development of such potentials has a long history, with early attempts to account for density effects dating back to the work of Dunn and Broyles¹⁷ and Lado,¹⁸ who introduced explicit density dependence to model many-body correlations. However, the approximations made while obtaining pseudopotentials may manifest themselves, especially at low temperatures. In particular, as shown by Jones and Murillo,¹⁹ the two-body approximation inherent in all pair pseudopotentials breaks down when many-body correlations become important. They demonstrated that even for weakly degenerate systems, the neglect of three-body and higher-order correlations leads to a failure of the pair approximation. This precisely corresponds to the regime where bound states and clusters emerge, limiting the applicability of any pairwise description.

Classical molecular dynamics simulations of dense semiclassical plasmas often employ effective ion-ion and electron-ion interactions derived within linear response or pseudopotential frameworks.^{14,16,20–24} To bridge the gap between such classical approaches and first-principles quantum simulations, hybrid methods have been proposed, in particular, the wave packet molecular dynamics combined with density functional theory.^{25–30} Another alternative is the electron force field approach³¹ that represents electrons as Gaussian wave packets with simplified antisymmetrization to capture Pauli repulsion efficiently. These approaches are frequently benchmarked against more fundamental theories, including first-principles simulations of dense hydrogen³² and quantum kinetic treatments of bremsstrahlung that highlight limitations of classical pseudopotentials in capturing high-frequency emission.³³ Recent advances in machine-learning potentials trained on Kohn–Sham density functional theory data now provide *ab initio*-quality reference data for validating classical models across a broad range of warm dense matter conditions,^{34,35} and transport-model intercomparisons further underscore the need for accurate effective interactions in strongly coupled regimes.³⁶

In this work, we study hydrogen classical TCP at relatively low temperatures using classical molecular dynamics with the standard¹⁰ and improved Kelbg¹⁴ pseudopotentials. In both cases, we observe the system collapse at $\Gamma \geq 1$ due to the unphysical clustering of electrons

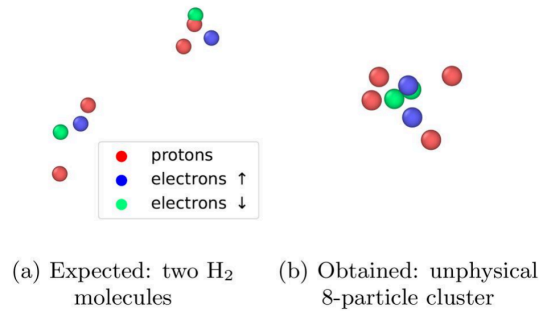


FIG. 1. Comparison of expected and obtained configurations for an eight-particle hydrogen system (four protons + four electrons) at $\Gamma = 1.0$. (a) Expected configuration: two hydrogen molecules (H_2), with electrons of identical spin projections spatially separated. (b) Obtained configuration: a single nonphysical cluster, reflecting the failure of the pairwise pseudopotential to reproduce fermionic exchange effects.

with the same spin projection. Figure 1 demonstrates this effect: while the left panel shows two properly bound hydrogen molecules formed with correct Pauli principle accounting, the right panel shows four protons and four electrons with identical spin projection collapsed into an unphysical dense cluster. This phenomenon can be clearly visible in the energy evolution and radial distribution functions (RDFs).

II. SIMULATION METHOD AND POTENTIALS

A. System setup

We consider a two-component hydrogen plasma consisting of N_p protons and N_e electrons, with $N_p = N_e = N/2$, confined in a cubic simulation cell of side length L with periodic boundary conditions. The system is charge-neutral, $\sum_{i=1}^N q_i = 0$, and contains equal numbers of electrons with spin projections $+\hbar/2$ and $-\hbar/2$, ensuring spin balance. All quantities are expressed in atomic units, where lengths are normalized to the mean interparticle distance $r_a = (4\pi n_e/3)^{-1/3}$, with $n_e = N_e/L^3$ being the electron density. Thermodynamic properties of the system can be expressed in terms of two dimensionless parameters: the coupling parameter Γ and the degeneracy parameter χ , defined as

$$\Gamma = \frac{e^2}{r_a k_B T}, \quad \chi = n_e \Lambda_e^3, \quad (1)$$

where $\Lambda_e = h/\sqrt{2\pi m_e k_B T}$ is the thermal de Broglie wavelength of an electron. The parameter Γ characterizes the strength of Coulomb interactions relative to thermal energy, while χ quantifies the importance of exchange effects and Fermi–Dirac statistics. The other common choice of parameters for the hydrogen plasma is the Brueckner parameter $r_s = r_a/a_B$ and temperature T . For the reader's convenience, we provide the conversion from the coupling (Γ) and degeneracy (χ) parameters used in our MD simulations

$$\frac{E_{Ha}}{k_B T} = (4.5\pi)^{1/3} \Gamma^2 \chi^{-2/3}, \quad r_s = (4.5\pi)^{1/3} \Gamma \chi^{-2/3}, \quad (2)$$

where E_{Ha} is the Hartree energy and a_B is the Bohr radius.

B. Interaction potentials and forces

The attractive Coulomb potential $-1/r$ makes the total energy of a TCP unbound from below. This leads to divergences in the configuration integral and unphysical collapse in simulations. To regularize the interaction at short distances and incorporate quantum effects, pseudopotentials such as the Kelbg pseudopotential¹⁰ are employed. Pseudopotentials are usually temperature-dependent and provide a finite value at zero distance.

The original Kelbg pseudopotential is derived from the diagonal element of the high-temperature two-particle density matrix and regularizes the Coulomb singularity at short distances by incorporating quantum diffraction effects through the thermal de Broglie wavelength $\lambda_{ij} = \sqrt{\hbar^2 \beta / (2\mu_{ij})}$, where $\beta = (k_B T)^{-1}$ and μ_{ij} is the reduced mass of the interacting pair

$$\Phi_0(r, \lambda_{ij}) = \frac{1}{r} \left[1 - e^{-\frac{r^2}{2\lambda_{ij}^2}} + \frac{\sqrt{\pi} r}{\lambda_{ij}} \operatorname{erfc}\left(\frac{r}{\lambda_{ij}}\right) \right]. \quad (3)$$

The improved Kelbg pseudopotential¹⁴ extends the applicability of the original model to lower temperatures by introducing a temperature-dependent correction parameter γ that ensures the correct asymptotic behavior of the potential at $r \rightarrow 0$:

$$\Phi_\gamma(r, \lambda_{ij}) = \frac{1}{r} \left[1 - e^{-\frac{r^2}{2\lambda_{ij}^2}} + \frac{\sqrt{\pi} \gamma_{ij} r}{\lambda_{ij}} \operatorname{erfc}\left(\frac{\gamma_{ij} r}{\lambda_{ij}}\right) \right]. \quad (4)$$

For electron–proton interaction, γ is defined as:

$$\gamma_{ep}(T) = \frac{x_1 + x_1^2}{1 + a_{ep}x_1 + x_1^2}, \quad x_1 = \sqrt{\frac{8\pi k_B T}{E_{\text{Ha}}}}, \quad (5)$$

where $a_{ep} = 1.090(14)$ and E_{Ha} is the Hartree energy. For the electron–electron interaction,

$$\gamma_{ee}(T) = \frac{\gamma_{ee}(T \rightarrow 0) + a_{ee}x_1 + x_1^2}{1 + x_1^2}, \quad (6)$$

where $a_{ee} = 0.18(1)$ and $\gamma_{ee}(T \rightarrow 0)$ is obtained from the zero-temperature limit as:

$$\gamma_{ee}(T \rightarrow 0) = -\frac{2}{\sqrt{\pi}} \frac{\tilde{x}^3}{\ln(8\tilde{x}^4/\sqrt{\pi}) - 3\tilde{x}^2}, \quad (7)$$

where $\tilde{x} = (\pi e^2 \beta / (2\lambda_{ee}))^{1/3}$. This correction preserves the correct slope $\partial \Phi_\gamma(r, \lambda_{ij}) / \partial r = -1/\lambda_{ij}^2$ at $r = 0$ and adjusts the potential depth, allowing for a more accurate description of atomic and molecular formation in partially ionized plasmas.

Subsequently, we compare two approaches: the standard Kelbg pseudopotential¹⁰ and the improved Kelbg pseudopotential¹⁴ augmented with spin-dependent exchange effects. The standard model employs the original Kelbg pseudopotential $\Phi_0(r, \lambda_{ij})$. In contrast, the improved model follows the methodology of Filinov *et al.*¹⁴ It utilizes a modified potential $\Phi_\gamma(r, \lambda_{ij})$ that incorporates a temperature-dependent fitting parameter $\gamma(T)$ to correct the potential value at $r = 0$, thereby reproducing the exact two-particle density matrix and accounting for bound states. Furthermore, for electron–electron interactions, we incorporate spin-dependent exchange effects by distinguishing between singlet and triplet states. It is important to note that

the improved Kelbg pseudopotential Φ_γ itself describes the interaction of distinguishable particles; exchange effects for identical fermions are incorporated separately via the antisymmetrization of the two-particle density matrix in the diagonal approximation.

Thus, while the standard model describes all pairwise interactions (ion–ion, electron–ion, and electron–electron) using $\Phi_0(r, \lambda_{ij})$, regardless of spin, the improved model employs a spin-resolved potential for electrons following Ref. 14. The total interaction potentials for the singlet (S) and triplet (T) states are given by:

$$\Phi(r, \lambda_{ij}) = \begin{cases} \Phi_\gamma(r, \lambda_{ij}), & \text{(a)} \\ \Phi_\gamma(r, \lambda_{ee}) - \frac{\ln(1 - e^{-r^2/\lambda_{ee}^2})}{\beta e^2}, & \text{(b)} \\ \Phi_\gamma(r, \lambda_{ee}) - \frac{\ln(1 + e^{-r^2/\lambda_{ee}^2})}{\beta e^2}. & \text{(c).} \end{cases} \quad (8)$$

In the above formula, (a) stands for ion–ion and ion–electron interaction, (b) for triplet electron interactions, and (c) for singlet ones. Here, the logarithmic terms arise from the antisymmetrization procedure: for triplet pairs, they emulate Pauli repulsion, while for singlet pairs, they allow enhanced correlation. For completeness, the proton–proton interaction is also modeled using the improved Kelbg pseudopotential. Although due to the large proton mass, this interaction could be treated classically with a bare Coulomb potential, we apply the quantum pseudopotential consistently across all species to ensure a uniform treatment of quantum diffraction effects.

In molecular dynamics, forces were computed as the gradient of the pseudopotential

$$\mathbf{F}_{ij} = -q_i q_j \nabla \Phi(r_{ij}, \beta). \quad (9)$$

To avoid numerical instabilities, the interaction force was smoothly truncated to zero at a cutoff radius $r = r_m$,³⁷ where

$$r_m = \left(\frac{3}{4\pi}\right)^{1/3} L \approx 0.62 L. \quad (10)$$

This radius corresponds to the radius of a sphere having the same volume as the cubic simulation cell of side L . In an isotropic medium, such a sphere encloses, on average, an electrically neutral region equivalent to the contents of the simulation box.³⁸ Moreover, r_m coincides with the interaction range of the angular-averaged Ewald potential for both one- and two-component plasmas.^{37,38} Due to the long-range nature of the Coulomb interaction, a finite truncation radius is necessary in classical simulations; the chosen value of r_m is sufficiently large to capture the essential physics of the interaction while ensuring numerical stability. Notably, at distances $r \gg \lambda_{ij}$, both the standard and improved Kelbg pseudopotentials asymptotically approach the form of the Coulomb potential. Thus, pseudopotentials inherently incorporate quantum diffraction and exchange effects at short range but decay similarly to the classical interaction at large separations.

The total energy of the system accounts for the temperature dependence of the pseudopotential

$$E = K + \sum_{i \neq j} \frac{q_i q_j}{2} \left[\Phi(r_{ij}, \beta) + \beta \left(\frac{\partial \Phi(r_{ij}, \beta)}{\partial \beta} \right)_v \right], \quad (11)$$

where $K = \sum_i m_i v_i^2 / 2$ is the kinetic energy of particles. All pseudo-potentials, pairwise forces, and energy contributions are calculated using the KelbgLIP code.³⁹

Although the Kelbg pseudopotential accounts for two-particle quantum effects—such as diffraction and the uncertainty principle—via the thermal de Broglie wavelength λ_{ip} , it does not automatically include many-body quantum statistics, such as exchange effects due to particle indistinguishability. This limitation restricts its applicability in classical molecular dynamics simulations to weakly degenerate regimes ($\chi \ll 1$), where many-body quantum statistical effects are negligible. In strongly degenerate systems ($\chi \gtrsim 1$), accurate modeling requires fully quantum approaches such as density functional theory-based molecular dynamics, path integral Monte Carlo (PIMC), or path integral molecular dynamics.

C. Statistical ensemble

The choice of the thermodynamic ensemble is critical for the accurate and efficient simulation of hydrogen plasma, particularly in regimes where the formation of bound states (hydrogen atoms and molecules) is possible. In this work, we employ the canonical (NVT) ensemble using the Nosé–Hoover thermostat scheme^{40,41} for the following reasons.

While the microcanonical (NVE) ensemble is suitable for simulating high-temperature plasmas where particles do not form bound states, it becomes computationally prohibitive at lower temperatures where bound electron–proton states emerge. The formation of atoms introduces high-frequency electron motion, necessitating an impractically small integration time step to maintain the conservation of total energy. This constraint renders NVE simulations of recombining plasmas inefficient for the long timescales required to study equilibrium properties.

The NVT ensemble circumvents this issue by maintaining a constant temperature, thus stabilizing the simulation and allowing for a viable time step without the stringent energy conservation requirement of NVE. To accurately represent the multi-component nature of the plasma, we apply three independent Nosé–Hoover thermostat chains to three distinct subsystems: protons, spin-up electrons, and spin-down electrons. This approach is essential for preventing the spurious transfer of thermal energy between disparate mass species via the thermostat itself, a known artifact when a single thermostat is applied to a system with large mass ratios.⁴²

The Nosé–Hoover thermostat was selected for its rigorous foundation in statistical mechanics, as it generates a dynamics that samples from the canonical ensemble. In the limit of a sufficiently large system, it produces the correct Boltzmann distribution of energies. This ensures that our simulation results for thermodynamic properties and population distributions are statistically meaningful.

D. Simulation parameters

Simulations were carried out using the LAMMPS package,⁴³ with initial particle positions randomized and velocities assigned according to the Maxwell–Boltzmann distribution.

A variable time step, ranging from 0.01 to 0.6 fs, was employed to maintain numerical stability, corresponding to the conditions of high ($T \sim 100$ kK, $\Gamma = 0.25$, and $\chi = 10^{-2}$) and low ($T \sim 1$ kK, $\Gamma = 2.5$, and $\chi = 10^{-2}$) temperature regimes, respectively.

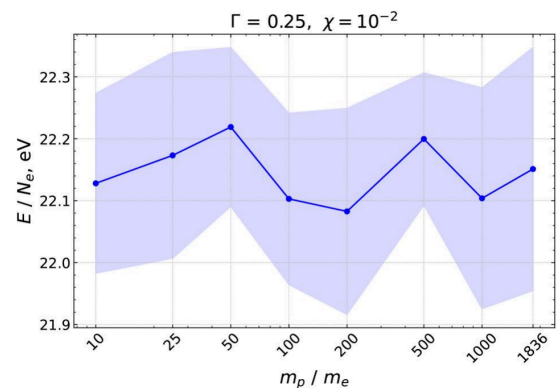


FIG. 2. Insensitivity of the equilibrium energy to the mass ratio. The energy per electron (E/N_e) is shown as a function of the proton-to-electron mass ratio. The error bars represent the standard deviation of the time-averaged energy from molecular dynamics simulations. The observed variation of 0.21% (relative standard deviation) confirms that the system's equilibrium energy is largely unaffected within the studied range, justifying the use of a reduced mass ratio for computational efficiency.

To accelerate the molecular dynamics calculations, an effective proton-to-electron mass ratio of $m_p/m_e = 100$ was adopted. As demonstrated in Fig. 2, the system's energy exhibits negligible dependence on the mass ratio within the range of 10–1836, with variations remaining within the statistical uncertainty of approximately 0.15 eV. This choice provides an 18-fold computational speedup compared to using the physical mass ratio. Mass ratios below 50 were avoided to prevent unphysical coupling between electron and ion dynamics, which would contradict the fact that protons are essentially stationary on the characteristic timescale of electron motion. Thus, the ratio of 100 represents an optimal compromise between computational efficiency and physical fidelity.

The model system consisted of 200 particles (100 protons, 50 spin-up electrons, and 50 spin-down electrons). This number represents a balance between computational tractability and physical accuracy, with key results—including energy and collapse behavior—being reproducible in larger systems.

The radial distribution functions (RDFs) were computed using the InterRDF module of MDAnalysis.^{44,45} The potential energy evolution was monitored to assess system stability, as well as an analysis of clustering in particle configurations.

III. RESULTS

A. Comparison of MD simulations using Kelbg, improved Kelbg, and Deutsch potentials with PIMC results

To validate the accuracy of the employed molecular dynamics models, we compare their predictions with benchmark results from path integral Monte Carlo (PIMC) simulations.⁴⁶ Figure 3 presents a comparison of the internal energy per electron, E/N_e , as a function of the Brueckner parameter r_s for two isotherms: $T = 95$ 250 and 62 500 K. The corresponding coupling and degeneracy parameters are listed in Table I. The results from the PIMC reference data (red circles) are plotted against those obtained using the standard Kelbg

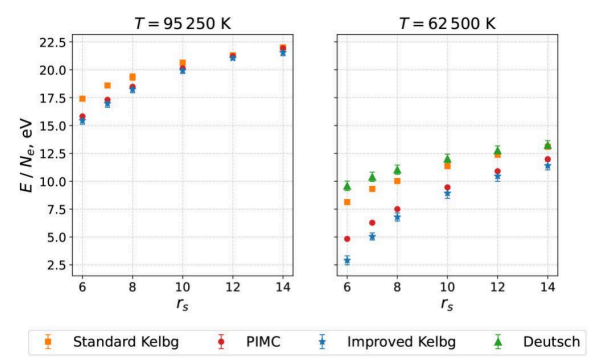


FIG. 3. Comparison of energies calculated using molecular dynamics with different potentials against PIMC reference data. Two isotherms are shown: $T=95\,250$ and $62\,500$ K. Red circles—PIMC data,⁴⁶ orange squares—standard Kelbg pseudopotential, and blue stars—improved Kelbg pseudopotential. The improved Kelbg pseudopotential demonstrates excellent agreement with PIMC at high temperature and good agreement in the $r_s > 10$ region at low temperature, while the standard Kelbg pseudopotential, being a high-temperature approximation, shows significant discrepancies at $T=62\,500$ K.

TABLE I. Correspondence between Brueckner parameter r_s , coupling parameter Γ , and degeneracy parameter χ for two isotherms used in PIMC validation (Fig. 3).

r_s	$T=95\,250$ K		$T=62\,500$ K	
	Γ	χ	Γ	χ
6	0.553	0.105	0.842	0.198
7	0.474	0.066	0.722	0.124
8	0.414	0.044	0.632	0.083
10	0.332	0.023	0.505	0.043
12	0.276	0.013	0.421	0.025
14	0.237	0.008	0.361	0.016

pseudopotential (orange squares) and improved Kelbg pseudopotential (blue stars).

At the higher temperature ($T=95\,250$ K), the improved Kelbg pseudopotential demonstrates excellent agreement with the PIMC benchmark across the entire range of r_s . The standard Kelbg pseudopotential shows minor deviations at $r_s < 10$, where coupling effects are stronger, but converges well with the reference data at higher r_s , where the system becomes less degenerate.

At the lower temperature ($T=62\,500$ K), the influence of quantum many-body effects that are fully accounted for in PIMC but not in MD with the pairwise Kelbg pseudopotentials, becomes more pronounced. For $r_s < 10$, both the standard and improved Kelbg pseudopotentials exhibit noticeable discrepancies from the PIMC results. However, for $r_s \geq 10$, the improved Kelbg pseudopotential recovers and shows good agreement with the benchmark. In contrast, the standard Kelbg pseudopotential, being a high-temperature approximation, fails to accurately describe the system at this temperature and deviates from the PIMC data for all values of r_s .

It is instructive to examine other available pseudopotentials, most of which are also high-temperature approximations. The

Deutsch pseudopotential^{12,16} [Eq. (2) in Ref. 47] was originally derived for a fully ionized plasma at temperatures above $k_B T > 1$ Ry. We performed MD simulations with this pseudopotential and plotted the results in Fig. 3 at $T=62\,500$ K. As can be seen, the high-temperature approximation fails at this temperature, yielding energies that deviate considerably from the PIMC benchmark. This is expected, as the Deutsch potential was not designed for the temperatures at which the influence of bound states is significant. Nevertheless, we performed simulations at $\Gamma = 1.0$ (corresponding to $T = 6061$ K) and observed no collapse—the system remained artificially nearly fully ionized, while molecular formation had to be substantial. This provides an additional argument for our focus on the improved Kelbg pseudopotential, as it is designed to capture the gradual onset of molecules at low temperatures.

B. Collapse investigation and pseudopotential artifacts

In this subsection, we focus on the weakly degenerate regime at a fixed degeneracy parameter $\chi = 10^{-2}$ across a wide range of coupling parameters $\Gamma = 0.25, 0.5, 1.0, 2.5$, corresponding to temperatures from $96\,986$ K down to 969 K.

The classical molecular dynamics simulations of hydrogen plasma were performed using both the standard and improved Kelbg pseudopotentials. The primary objective was to assess the stability and physical consistency of these two approaches in the regime where quantum exchange effects become significant. Both pseudopotentials reproduce the Debye–Hückel limit within the computational accuracy.³⁹ The computations were performed for up to 10^6 particles at $\Gamma = 0.01$ using a classical Monte Carlo method and path integral Monte Carlo. It was shown that one can neglect long-range interaction effects at weak interaction.³⁷

At $\Gamma = 0.25$ ($T=96\,986$ K), both the standard and improved Kelbg pseudopotentials produce physically reasonable results: the system consists mostly of free protons and electrons, with a very small fraction of bound states formed (Fig. 4). This is consistent with the

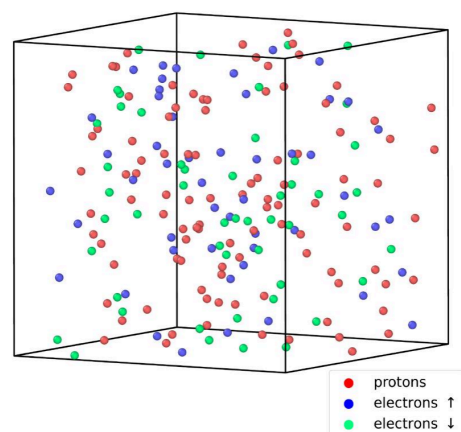


FIG. 4. Snapshot of the computational cell on the equilibrium part of a simulation. The system contains no visible bound states, $\Gamma = 0.25$.

expected behavior of a weakly coupled, nearly ideal plasma below the ionization threshold.

As Γ increases to 0.5 ($T = 24\,246\text{ K}$), atomic hydrogen begins to form, signaling the onset of bound states. The standard Kelbg pseudopotential produces only stable atomic hydrogen configurations [Fig. 5(a)] without molecular formation. In contrast, the improved Kelbg pseudopotential demonstrates enhanced capability by reproducing both atomic configurations [Fig. 5(a)] and correct H_2 molecules [Fig. 5(b)].

However, this improvement comes with critical limitations. The improved pseudopotential begins to exhibit breakdown of the exchange approximation, manifesting as unphysical configurations including incorrect H_2 molecules where both electrons possess identical spin projections [Fig. 5(c)], and the formation of small electron-proton clusters containing multiple electrons with parallel spins [Fig. 5(d)]. These artifacts indicate the breakdown of the spin-dependent pseudopotential at this coupling strength, allowing electrons of identical spin to occupy overlapping spatial regions. Despite these unphysical features, the system remains numerically stable at $\Gamma = 0.5$.

At $\Gamma = 0.5$, the proton-proton radial distribution functions (Fig. 6) reveal a clear distinction between the two pseudopotentials. While the standard Kelbg pseudopotential yields a monotonically increasing $g_{pp}(r)$ that starts from zero and asymptotically approaches unity—consistent with a plasma of free protons and isolated hydrogen

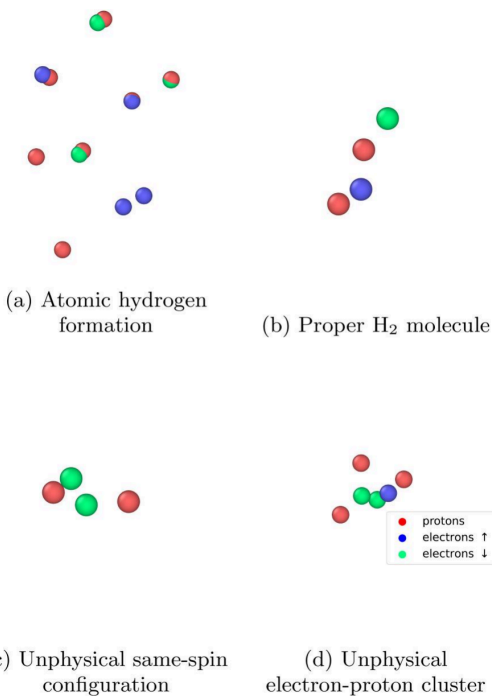


FIG. 5. Configurations observed at $\Gamma = 0.5$: (a) atomic hydrogen; (b) correct H_2 molecule with antiparallel spins; (c) unphysical same-spin configuration; and (d) three-particle cluster with spin polarization artifacts.

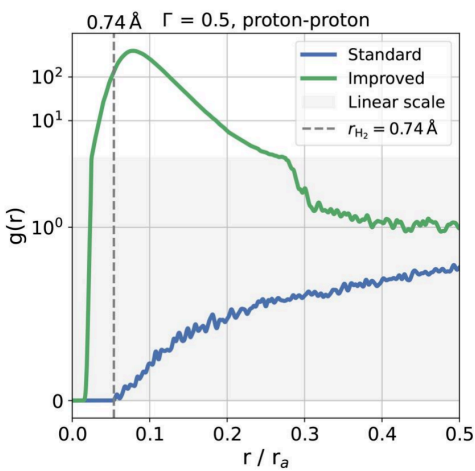


FIG. 6. Proton-proton radial distribution functions at $\Gamma = 0.5$ for the standard and improved Kelbg pseudopotentials. The peak in the improved case near $r \approx 0.74\text{ \AA}$ indicates the presence of H_2 -like structures and incipient clusters.

atoms—the improved Kelbg pseudopotential exhibits a pronounced peak at short range. This peak corresponds to $r \approx 0.74\text{ \AA}$ in physical units (the equilibrium bond length in H_2), signaling the formation of multi-proton bound complexes. As discussed earlier, these include genuine H_2 molecules as well as incipient unphysical clusters involving electrons with parallel spins. The slight shift of the peak relative to the ideal H_2 bond length likely reflects the contribution of these non-molecular aggregates, which distort the average interproton separation.

The situation becomes catastrophic at $\Gamma = 1.0$ ($T = 6061\text{ K}$) and $\Gamma = 2.5$ ($T = 969\text{ K}$). Both the standard and improved Kelbg pseudopotentials lead to a complete system collapse (Fig. 7). The electrons form a dense central cluster, while the protons are pushed to the periphery. This configuration is fundamentally unphysical and forbidden by quantum mechanics, as it represents a gross failure of the pairwise pseudopotential to reproduce fermionic exchange effects.

C. Quantitative signatures of pseudopotential failure

The qualitative observations of unphysical clustering find their quantitative confirmation in both structural and energetic properties

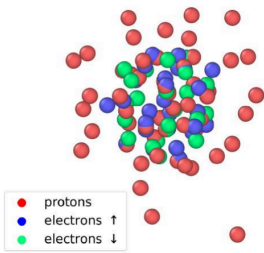


FIG. 7. Full collapse, $\Gamma = 1.0$ and 2.5 .

of the system. The radial distribution functions (RDFs) for electrons with identical spin projections provide the clearest structural evidence of pseudopotential failure. At weak coupling ($\Gamma = 0.25$ and 0.5 for standard Kelbg, and $\Gamma = 0.25$ for improved Kelbg), the RDF $g(r)$ properly approaches zero as $r \rightarrow 0$, reflecting the expected quantum behavior due to fermionic exchange effects. However, for $\Gamma \geq 1.0$, both pseudopotentials exhibit a pronounced peak at $r < \lambda_{ee}$, clearly visible in Figs. 8(a) and 8(c). This peak represents a breakdown of the pairwise exchange approximation, indicating a high probability of finding two electrons with identical spins at distances less than their localization region.

The dynamical signature of this failure is captured by the time evolution of the potential energy. For $\Gamma = 1.0$ and 2.5 , both pseudopotentials exhibit a characteristic step-like decrease in energy over time [Figs. 8(b) and 8(d)], with no indication of reaching thermodynamic equilibrium. Each discrete energy drop corresponds to the formation of another unphysical cluster, revealing that the system undergoes progressive collapse rather than relaxation to a stable state. This behavior is particularly concerning as it demonstrates that the improved Kelbg pseudopotential, despite its theoretical enhancements

TABLE II. Applicability limits of Kelbg pseudopotentials. System configurations, radial distribution function $g(r)$ for same-spin electrons, and simulation stability are shown for different coupling strengths Γ . Corresponding temperatures are: $\Gamma = 0.25$ ($T \approx 97$ kK), $\Gamma = 0.5$ ($T \approx 24$ kK), $\Gamma = 1.0$ ($T \approx 6$ kK), and $\Gamma = 2.5$ ($T \approx 1$ kK).

Γ	Configuration	$g(r)$ behavior	Stability
0.25	Standard: Free particles predominate	Monotonic	Stable
	Improved: Free particles predominate	Monotonic	Stable
0.5	Standard: Atomic H	Monotonic	Stable
	Improved: Atomic H, molecules, clustering	Small peak	Stable, artifacts
1.0	Standard: Collapsed	Pronounced peak	Unstable
	Improved: Collapsed	Pronounced peak	Unstable
2.5	Standard: Collapsed	Pronounced peak	Unstable
	Improved: Collapsed	Pronounced peak	Unstable

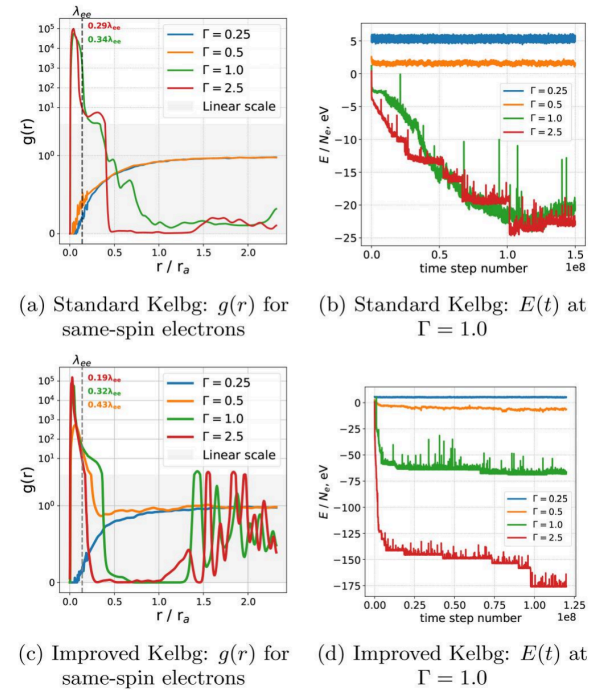


FIG. 8. Comparative failure of the standard and improved Kelbg pseudopotentials at strong coupling. (a) Standard Kelbg: emergence of unphysical peak in $g(r)$ at $r < \lambda_{ee}$ for $\Gamma \geq 1.0$, indicating Pauli principle violation. Colored annotations mark the peak positions. (b) Corresponding total energy shows step-like decrease, confirming absence of stable equilibrium. (c) Improved Kelbg exhibits identical pathological peak in $g(r)$ despite its strong-coupling corrections, with analogous peak positions. (d) Energy evolution demonstrates similar instability through discontinuous drops. Both pseudopotentials fundamentally fail to reproduce fermionic exchange effects at $\Gamma \geq 1.0$.

for strong coupling, ultimately suffers from the same fundamental limitation as its standard counterpart—the inability to maintain proper quantum statistics under conditions of increased degeneracy and coupling strength.

A comprehensive summary of these findings is presented in Table II. It establishes that both the standard and improved Kelbg pseudopotentials are stable and produce physically meaningful results only for $\Gamma < 0.5$. In this regime, the system is well-described as a classical plasma with weak correlations. However, for $\Gamma > 0.5$, both methods fail due to unphysical clustering, rendering them unreliable for simulations of dense, partially ionized hydrogen plasmas at low temperatures.

IV. DISCUSSION AND CONCLUSION

Our systematic investigation establishes clear applicability boundaries for classical molecular dynamics with quantum pseudopotentials in modeling hydrogen plasma. The results demonstrate that existing pseudopotentials, including the improved Kelbg formulation, become fundamentally inadequate for systems with $\Gamma > 0.5$, corresponding to temperatures below approximately 25 kK.

The failure manifests itself through distinct quantitative signatures: unphysical peaks in the radial distribution function for same-spin electrons at distances $r < \lambda_{ee}$, and step-like decreases in potential energy indicating progressive system collapse. These artifacts stem from the inability of the considered pairwise pseudopotentials to enforce the minimum approach distance required by the Pauli exclusion principle.

Within the valid regime ($\Gamma \leq 0.5$), the improved Kelbg pseudopotential shows superior performance compared to the standard formulation, particularly at intermediate coupling strengths, where it successfully reproduces both atomic and molecular configurations while maintaining agreement with PIMC benchmarks. However, its advantage diminishes as the system approaches the collapse boundary, where both potentials exhibit similar failures.

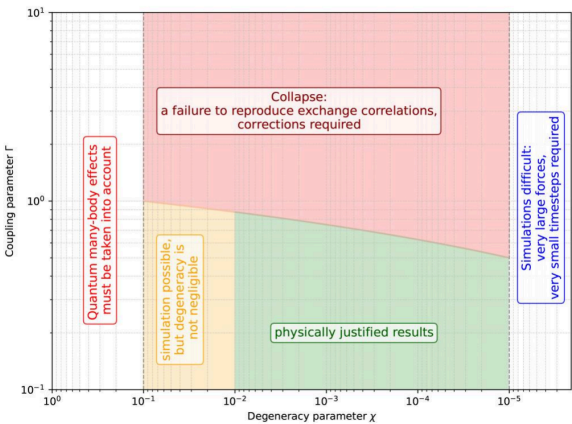


FIG. 9. Regions of applicability of molecular dynamics with the improved Kelbg pseudopotential in the coupling-degeneracy (Γ - χ) plane.

Figure 9 summarizes the operational domain of MD with the improved Kelbg pseudopotential, bounded by three fundamental constraints: the onset of significant quantum many-body effects at $\chi > 0.1$, numerical difficulties at low degeneracy ($\chi < 10^{-4}$), and the physical breakdown due to the failure of the pairwise pseudopotential to reproduce fermionic exchange effects at $\Gamma > 0.5$. This work, therefore, establishes both the capabilities and limitations of current classical approaches for quantum plasma simulation and highlights the necessity for developing pseudopotentials with explicit spin-dependent repulsive cores or alternative computational strategies for the strongly coupled regime.

In addition to equilibrium properties, nonequilibrium effects, in particular, electron-ion energy relaxation, are of clear physical interest.^{48–51} At the same time, such processes are difficult to address consistently within a framework based on temperature-dependent pseudopotentials, because the interaction itself changes with temperature. Approximate schemes have been discussed in the literature, for example, in Ref. 52, where the pseudopotential is kept fixed at a reference temperature while a weak temperature imbalance between electrons and ions is introduced. However, constructing a rigorous methodology for nonequilibrium simulations with temperature-dependent pseudopotentials remains a challenging problem and lies beyond the scope of the present study.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

A. S. Onegin: Conceptualization (equal); Data curation (lead); Formal analysis (equal); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead). G. S. Demyanov: Conceptualization (equal); Methodology (equal). P. R. Levashov: Conceptualization (equal); Funding acquisition (lead); Methodology (equal); Project administration (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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